

A32NX weather radar on the ND

A native weather radar picture on both A32NX NDs, from MSFS's own radar data

Aircraft	A32NX
Where to find it	ND in ARC or ROSE NAV; the weather radar panel on the pedestal (WX SYS and MODE knobs)
PR branch	pr/01-a32nx-nd-weather-radar (base of PRs 2, 4, 3, 9)
Status	Tested in the sim over several sessions and weather presets (September 2026).
Pictures	In-sim screenshots taken during development (the radar is drawn by a native WASM gauge, which the browser harness cannot run).

1. What it is for

2. Using it

3. Reading the picture
4. What to expect

5. Limitations and why

1. What it is for

The A32NX ND had no working weather radar. A native gauge (`ndwxr`) now draws the MSFS weather radar on the CPT and F/O NDs, aligned with the range rings, in the colours of an airborne radar.



Figure 1. A32NX ND in ARC with weather ahead: green, yellow and red cells.

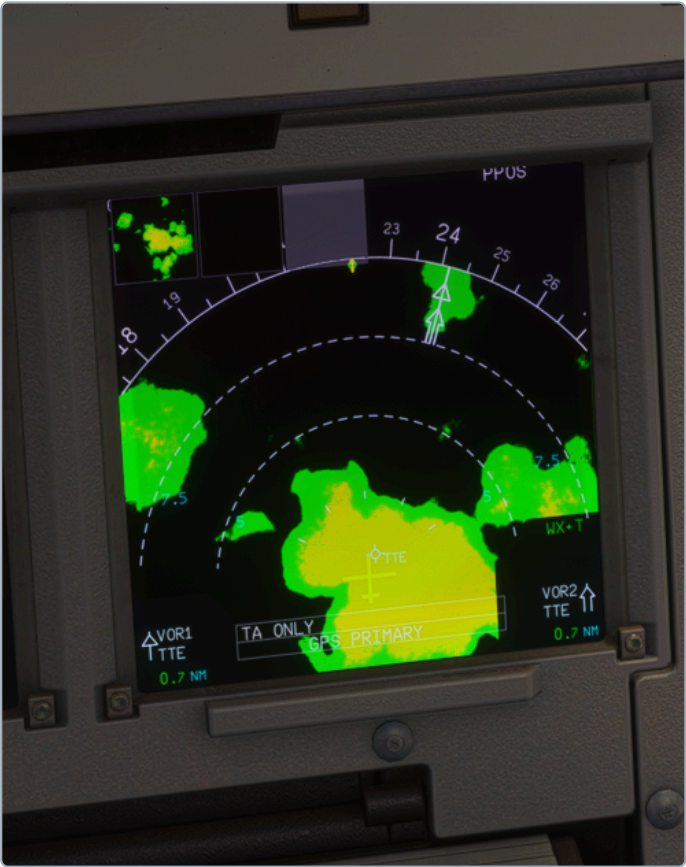


Figure 2. Another cell field, ROSE-like forward view.

2. Using it

Control	Position	ND
WX SYS knob	SYS 1 / SYS 2	Radar on (system 1 or 2)
	OFF	No radar

MODE knob	WX	Precipitation
	WX+T	Precipitation and turbulence (magenta)
	TURB	Turbulence only
	MAP	Nothing (no ground map)
EFIS	ARC or ROSE NAV	Radar shown (forward half); nothing on ROSE ILS, ROSE VOR, PLAN

The selected mode is written in green on the ND (see the PR 3 manual).

3. Reading the picture

Colour	Rain rate	Meaning
Green	from 0.7 mm/h	Light precipitation
Yellow	from 4 mm/h	Moderate
Red	from 12 mm/h	Heavy: avoid
Magenta	above 20 mm/h, within 40 NM	Turbulence (WX+T, TURB)

The engine's rain rate is noisy pixel by pixel; each level is smoothed, so the picture shows solid areas with crisp edges.

4. What to expect

Situation	What you should see
On the ground	Nothing drawn: like a real radar, it does not transmit with both main gear compressed.
Clear sky	No picture (nothing to return).
WX SYS OFF, or MODE MAP	No picture.
CPT and F/O NDs	Independent: each ND uses its own IR position (IR 1 CPT, IR 2 F/O, IR 3 by the ATT HDG knob) and its own range and mode; the two knobs are shared.

5. Limitations and why

Limitation	Why	What to do
Turbulence is a proxy.	MSFS gives no turbulence data; rain above 20 mm/h within 40 NM is drawn magenta.	-
No tilt, no gain.	The MSFS radar has neither.	-
Drawn only airborne with weather around.	The MSFS radar image needs both.	Test with the MSFS weather presets.